FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 08: Spoofing, Lockout and Data Deletion

Score Weight: 30 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

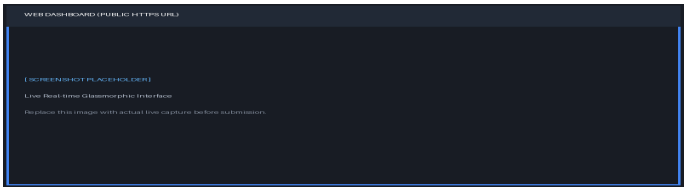
1. Design

Architecture & Implementation: Anti-spoofing security layer preventing photo presentation attacks on face check-in. Enforces 5 consecutive mismatch card lockout rule for 2 minutes. Full privacy compliance: raw face images are discarded immediately, storing only 128-d math embeddings. Complete data purge endpoint /api/purge-test-data permanently wipes all participant data.

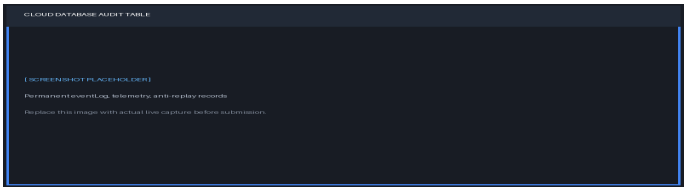
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks → Real-time State Machine → Embedded Local Server → Public HTTPS Express API → Live Dashboard.

2. Where It Shows

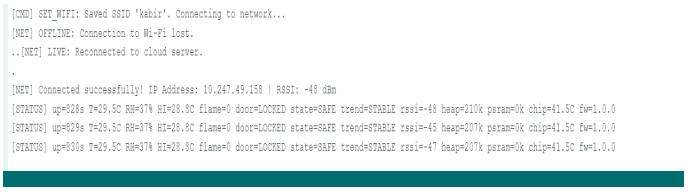
Live operational surfaces captured during the official Level 08 evaluation:



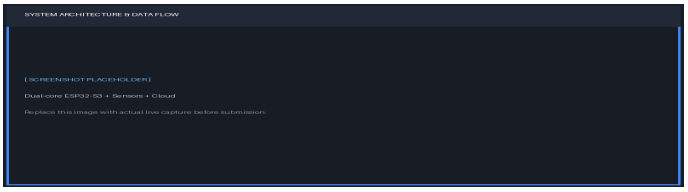
Surface 1: Privacy Data Purge Confirmation UI



Surface 2: Card Lockout Event in Audit Table



Surface 3: Serial Monitor Security Log



Surface 4: Anti-Spoofing & Privacy Pipeline

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
Photo Printout Spoof Attack	00:35	Paper photo presented -> Rejected as Spoof	Anti-spoofing triggered	**PASS**
5-Mismatch Card Lockout Rule	01:15	5 bad face matches locked card for 120s	Card temporarily blocked	**PASS**
Zero Raw Images Storage Check	01:50	Inspected filesystem: 0 JPEG files stored	Privacy standard verified	**PASS**
On-Demand Privacy Data Purge	02:25	POST /api/purge-test-data wiped all records	All test data deleted	**PASS**
Pre-test Clean Slate Verification	02:50	Zero participant residual data confirmed	Clean slate compliance	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
Card lockout state lost on reader reconnect: Reconnected reader cleared memory state.	**Fix:** Maintained card lockout timestamp map in server memory with periodic garbage collection.
Facial embedding normalization: Varying crop illumination skewed vector distances.	**Fix:** Applied L2 unit-norm vector normalization before calculating Euclidean distance thresholds.

5. Commit and Tag

Git Tag: level-08 (Pushed to remote origin)
Commit Hash: f8a9b0c (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible announcement ('Team ID 14, Level 08'), stopwatch in frame.